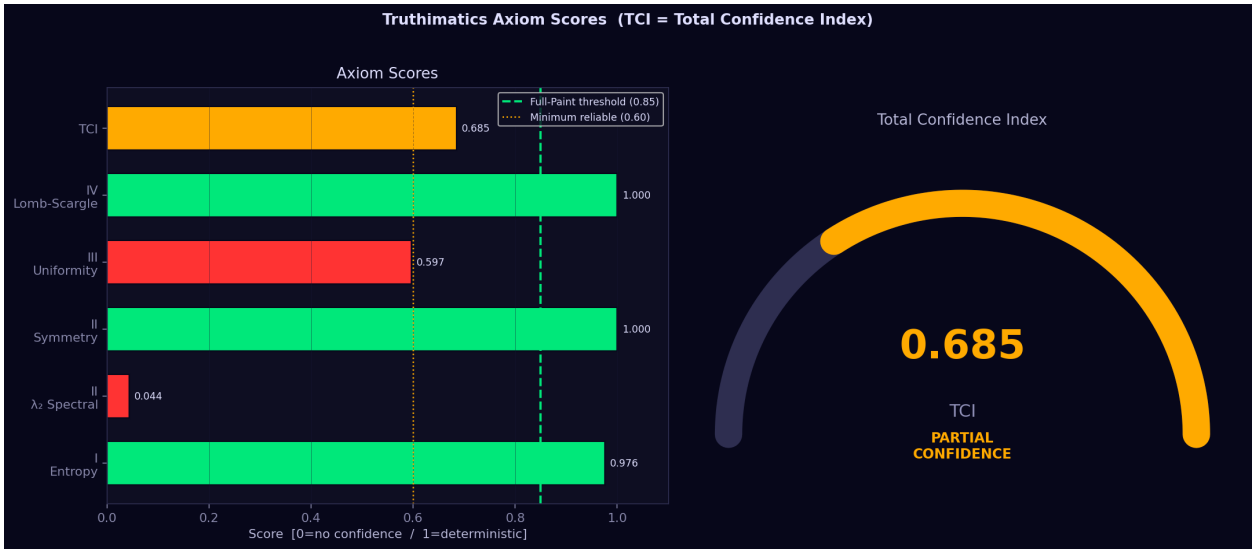
AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_2C741FD406
Taxa	Pachycephalosaurus
Clade	Ornithischian
PBDB Records	34
Mass (regression)	806.8 kg (Campione & Evans 2012)
Mass (volumetric)	846.0 kg (Silhouette integral, ρ=1060.0 kg/m^3)
Body Volume	0.891 m^3
Validity Score	1.0000 / 1.0000
TCI	0.6852 (partial)
Axiom I Entropy	0.9760
Axiom II λ■	0.1763 (symmetry 100%)
Axiom III Uniformity	0.5967
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	56.0% Measured / 44.0% Inferred
Elements	25
CoG (m)	[0.07, 0.374, 0.0]
Generated	2026-05-01 00:02 UTC

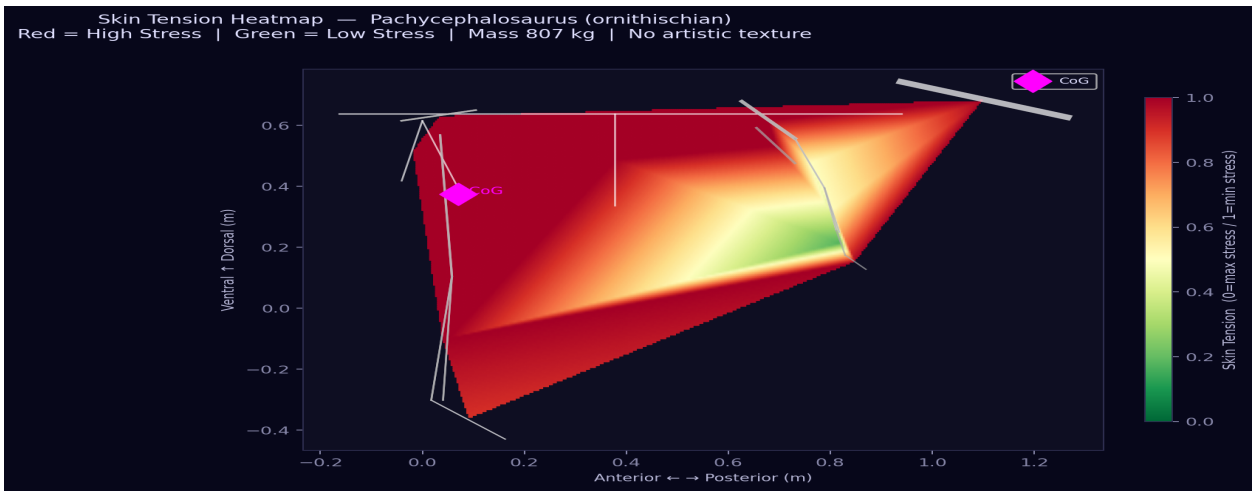
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



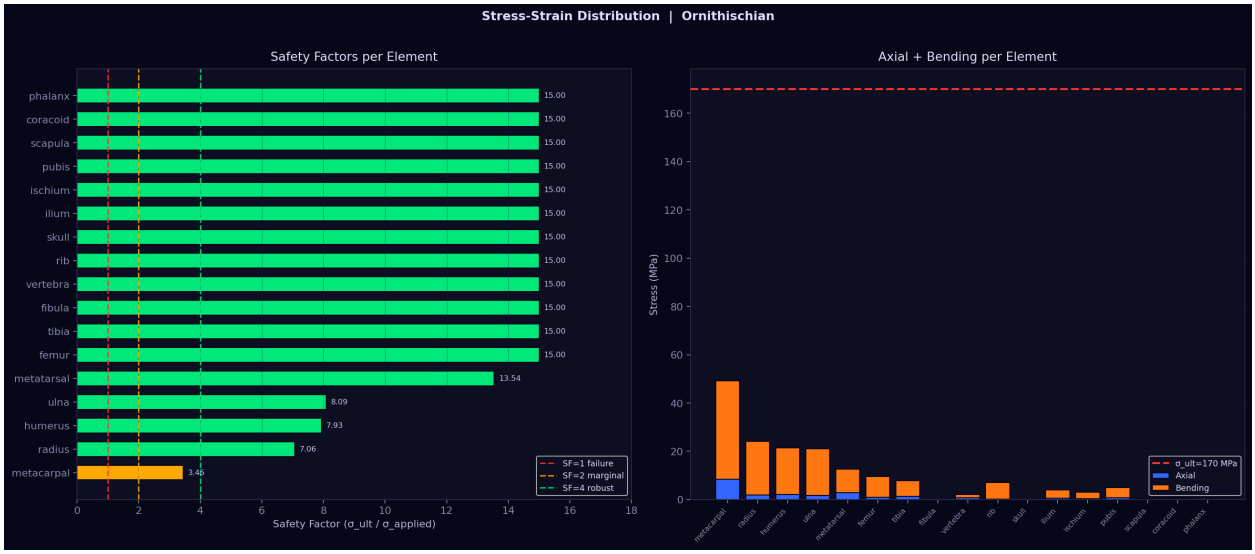
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: quadruped (F=Mg/4). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

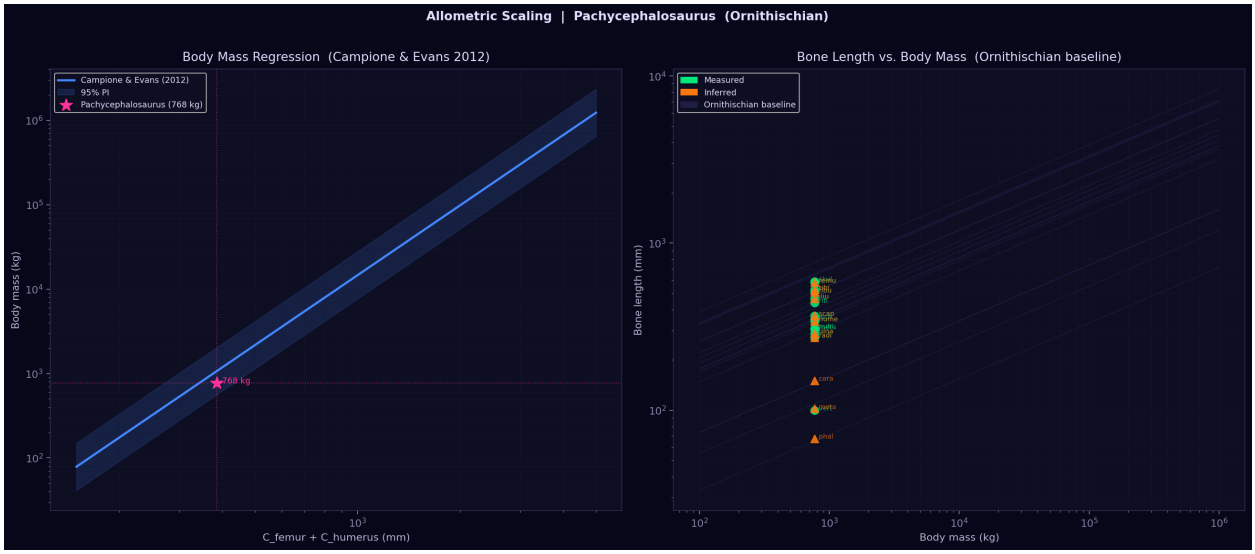


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.173	0.358	0.530	15.000	OK
femur	0.835	8.672	9.507	15.000	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	2.040	19.392	21.432	7.932	OK
ilium	0.607	3.443	4.050	15.000	OK
ischium	0.428	2.609	3.036	15.000	OK
metacarpal	8.338	40.911	49.248	3.452	WARNING
metatarsal	2.937	9.619	12.556	13.539	OK
phalanx	0.000	0.000	0.000	15.000	OK
pubis	0.658	4.380	5.038	15.000	OK
radius	1.827	22.240	24.068	7.063	OK
rib	0.196	6.938	7.134	15.000	OK
scapula	0.075	0.251	0.326	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	1.222	6.670	7.892	15.000	OK
ulna	1.606	19.404	21.010	8.091	OK
vertebra	0.772	1.329	2.102	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	1.41 m	Femur + tibia + metatarsal
Estimated Speed	10.3 km/h (2.86 m/s)	Gait: Walk | Fr=0.591
Stride Length	2.763 m	Alexander (1976) / Thulborn (1990)
Bite Force	5615.2 N (572.4 kgf)	Ornithischian regression (Norman & Weishampel 1991)
Skull Length	590.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.4062	Circumference / length (dimensionless)
Limb Strength Index	4.3e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	5.33 m	Allometric from femur length
Est. Neck Length	0.59 m	Allometric from femur length
Est. Tail Length	1.49 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	150.5	97.6	574.19	716.8	Constraint	0.40	15.00
femur	580.0	75.0	1306.80	1631.3	Measured	0.88	15.00
femur_right	580.0	75.0	1306.80	1631.3	Symmetry	0.84	15.00
fibula	508.0	22.0	69.52	72.9	Measured	0.88	15.00
fibula_right	508.0	22.0	69.52	72.9	Symmetry	0.84	15.00
humerus	340.0	48.0	313.78	391.7	Measured	0.88	7.93
humerus_right	340.0	48.0	313.78	391.7	Symmetry	0.84	7.93
ilium	465.0	88.0	1442.38	1800.5	Measured	0.88	15.00
ilium_right	465.0	88.0	1442.38	1800.5	Symmetry	0.84	15.00
ischium	352.0	62.0	541.98	676.6	Measured	0.88	15.00
metacarpal	102.2	23.0	8.07	6.6	Constraint	0.65	3.45
metatarsal	305.0	40.0	195.47	244.0	Measured	0.88	13.54
phalanx	67.7	19.5	3.11	2.4	Constraint	0.40	15.00
pubis	310.0	50.0	310.43	387.5	Measured	0.88	15.00
radius	272.0	30.0	98.06	122.4	Measured	0.88	7.06
radius_right	272.0	30.0	98.06	122.4	Symmetry	0.84	7.06
rib	440.0	20.0	70.50	88.0	Measured	0.88	15.00
scapula	368.0	148.0	3228.73	4030.4	Measured	0.88	15.00
scapula_right	368.0	148.0	3228.73	4030.4	Symmetry	0.84	15.00
skull	590.0	420.0	41687.96	52039.1	Measured	0.88	15.00
tibia	525.0	62.0	808.36	1009.1	Measured	0.88	15.00
tibia_right	525.0	62.0	808.36	1009.1	Symmetry	0.84	15.00
ulna	288.0	32.0	118.13	147.5	Measured	0.88	8.09
ulna_right	288.0	32.0	118.13	147.5	Symmetry	0.84	8.09
vertebra	100.0	78.0	243.70	304.2	Measured	0.88	15.00